

ANR ROBINSON INTL, Trinidad and Tobago

WMO: 789613

Lat: 11.150N

Lon: 60.833W

Elev: 12

StdP: 101.18

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00410

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.8	23.0	18.1	13.0	25.5	18.8	13.7	25.4	10.9	29.0	10.1	28.8	1.4	60	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	5.3	32.0	26.1	31.4	25.9	31.2	25.8	27.1	30.7	26.7	30.3	26.6	30.2	6.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	29.7	25.8	21.1	29.4	25.2	20.4	28.9	85.5	30.7	84.0	30.7	82.9	30.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.8	8.2	DB	20.7	33.1	1.8	0.6	19.4	33.6	18.3	33.9	17.2	34.3	15.9	34.8	(4)
(5)			WB	18.8	27.9	1.4	0.6	17.8	28.3	17.0	28.6	16.2	28.9	15.3	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.5	26.5	27.0	27.6	28.1	27.7	27.7	28.0	28.3	28.0	27.5	27.0	(6)	
	DBStd	1.03	0.71	0.80	0.88	0.81	0.79	0.88	0.81	0.89	0.88	0.91	0.89	0.77	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6392	512	462	527	529	561	532	550	558	551	558	526	526	(10)	
	CDD18.3	3350	253	229	268	279	302	282	292	300	301	300	276	268	(11)	
	CDH23.3	34057	2081	1881	2461	2877	3316	3023	3102	3351	3481	3303	2769	2412	(12)	
	CDH26.7	10045	491	446	645	845	1043	869	913	1104	1243	1073	784	588	(13)	
(14) Wind	WSAvg	3.8	3.6	4.0	4.3	4.7	5.0	4.5	3.5	3.0	3.3	3.3	3.2	3.2	(14)	
(15) Precipitation	PrecAvg	1452	46	42	45	39	59	143	178	158	166	218	219	149	(15)	
	PrecMax	1762	142	115	139	89	194	286	341	248	274	385	496	315	(16)	
	PrecMin	836	7	3	5	4	9	40	75	61	54	101	95	34	(17)	
	PrecStd	232	36	29	38	28	42	72	70	54	72	88	96	81	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	30.9	31.2	31.4	32.1	31.8	31.6	32.2	32.2	32.2	31.6	31.1	(19)	
		MCWB	24.7	24.7	24.7	25.3	25.6	25.7	25.8	26.3	26.4	26.4	26.2	25.7	(20)	
	2%	DB	30.1	30.1	30.7	31.1	31.2	31.1	31.1	31.7	31.9	31.4	31.1	30.5	(21)	
		MCWB	24.4	24.3	24.5	25.1	25.4	25.6	25.8	26.2	26.4	26.3	26.2	25.4	(22)	
	5%	DB	29.8	29.8	30.1	30.6	31.0	30.8	30.8	31.1	31.2	31.1	30.8	30.0	(23)	
		MCWB	24.3	24.1	24.2	24.9	25.3	25.5	25.7	26.1	26.1	26.2	26.1	25.1	(24)	
	10%	DB	29.1	29.1	29.8	30.1	30.2	30.1	30.1	30.8	31.0	30.8	30.1	29.4	(25)	
		MCWB	23.8	23.6	24.1	24.7	25.0	25.3	25.4	25.9	26.0	26.1	25.7	24.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	25.7	26.1	26.6	26.7	26.9	27.0	27.2	27.5	27.2	27.1	26.7	(27)	
		MCDB	29.5	29.4	29.8	30.4	30.4	30.3	30.4	30.9	30.9	30.8	30.5	30.1	(28)	
	2%	WB	25.2	25.1	25.4	26.0	26.4	26.5	26.6	26.7	26.9	26.8	26.7	26.2	(29)	
		MCDB	29.1	29.1	29.3	29.9	30.0	29.8	30.0	30.4	30.7	30.5	30.1	29.5	(30)	
	5%	WB	24.7	24.5	24.9	25.6	26.0	26.0	26.2	26.5	26.6	26.6	26.4	25.7	(31)	
		MCDB	28.5	28.6	28.9	29.4	29.6	29.4	29.7	30.2	30.4	30.1	29.8	28.9	(32)	
	10%	WB	24.3	24.1	24.4	25.1	25.6	25.7	25.8	26.2	26.3	26.2	26.0	25.2	(33)	
		MCDB	28.0	28.1	28.5	28.9	29.2	28.9	29.2	29.7	30.0	29.7	29.3	28.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	5.9	5.7	5.4	5.2	5.0	5.2	5.3	5.3	5.3	5.4	5.5	(35)	
		MCDBR	6.3	6.3	6.0	5.7	5.6	5.5	5.6	5.8	5.6	5.7	5.9	5.9	(36)	
	5% WB	MCWBR	3.0	3.0	2.8	2.6	2.5	2.6	2.8	2.8	2.8	2.8	3.0	3.0	(37)	
		MCWBR	5.8	5.9	5.7	5.3	5.1	5.1	5.3	5.5	5.4	5.5	5.6	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.383	0.405	0.400	0.448	0.470	0.524	0.491	0.439	0.417	0.392	0.383	0.395	(40)	
		taud	2.578	2.486	2.518	2.382	2.311	2.129	2.248	2.455	2.512	2.612	2.630	2.567	(41)	
	Edn at Noon	Edn at Noon	924	913	919	866	831	777	806	861	890	914	918	903	(42)	
		Edh at Noon	99	112	110	125	132	156	139	115	109	97	93	98	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.30	5.95	6.28	6.42	6.09	5.67	5.88	6.05	6.05	5.46	4.95	4.89	(44)	
	RadStd	RadStd	0.27	0.25	0.30	0.26	0.40	0.28	0.34	0.26	0.28	0.24	0.34	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.20	+0.33	+0.41	+0.14	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page